

Subject Description Form

Subject Code	ITC3071K
Subject Title	Coloration and Finishing for Knitwear
Credit Value	3 credits
Level	3
Pre-requisite/ <Co-requisite> / (Exclusion)	Nil
Objectives	This subject aims to provide students with an in-depth knowledge of the principles of textile preparatory processes and develop the students' understanding on the basic concept and technology of textile coloration and finishing for knitwear products.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand and describe the capabilities and limitations of the various preparation, dyeing and finishing processes in specific applications. - Apply their knowledge of mechanisms and technologies related to textile preparation, dyeing and finishing (i.e. wet processing), to select the correct machinery, processes, processing conditions and technologies to achieve the specific effects required for different end-uses and to minimise and solve technical problems, including environmental ones. - Describe and compare the influence of various processing conditions on the properties of the finished products.
Subject Synopsis/ Indicative Syllabus	(I) Preparation of Textiles Basic principles, mechanism and technology of preparatory process including scouring and bleaching processes for textile materials. Preparation of textile materials in various physical forms including knitwear garment using different types of machinery. (II) Coloration of Textiles Basic principles, mechanism and technology of dyeing process for textile materials. Dyeing of textile materials in the form of fibre, yarn, knitted fabric and knitwear garment using different types of dyeing machines. Classification of dyes according to application methods and the colour fastness requirements of dyed goods. Basic principles, mechanism, technology and styles of textile printing for textile materials including fixation and washing-off of prints. Printing of textile materials in the form of fibre, yarn, knitted fabric and knitwear garment using different types of printing machines, e.g. screen printing machine and digital printing machine. (III) Finishing of Textiles Basic principles, mechanism and technology of finishing process to impart dimensional stability, aesthetic and functional effects to textile materials. Finishing of textile materials in various physical forms including knitwear garment using different types of machinery and additives.

Teaching/Learning Methodology	Lectures are used to provide an overview of the textile preparation, coloration and finishing processes, and develop students’ understanding of the basic principle, mechanism and technology involved. Laboratory experiments are designed to provide students with hands-on experience to deepen their understanding of the concepts related to various textile preparation, coloration and finishing processes, and enhance their practical skills of using different wet processing techniques. Case analyses on technical problems and mini-projects of problem-solving nature will be assigned to students to develop their problem solving skills.																																													
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th></th><th></th><th></th></tr><tr><td>1. Coursework</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>2. Final Examination</td><td>50%</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: This subject provides the students with fundamental theory and some basic laboratory training so that the students can easily understand the preparation, coloration and finishing of knitwear products. Course works such as laboratory reports, group projects and lectures are used to assess the student’s learning outcomes at different stages and aspects of the subject. Examination is used to assess students’ intellectual competence in performing textile wet processing, analysing and solving problems as well as understanding the overall theory of the subject.								Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c				1. Coursework	50%	✓	✓	✓				2. Final Examination	50%	✓	✓	✓				Total	100%						
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Reading List and References	Books Karmakar, S. R. (1999). <i>Chemical Technology in the Pre-treatment Processes of Textiles</i>. Amsterdam; New York : Elsevier. Choudhury, A.K.R. (2006). <i>Textile Preparation and Dyeing</i>. Enfield, N.H.: Science Publishers. Park, J. and Shore, J. (2004). <i>Practical Dyeing</i>. Bradford, England: The Society of Dyers and Colourists. Broadbent, A.D. (2001). <i>Basic Principles of Textile Coloration</i>. Bradford: The Society of Dyers and Colorists.																																													

	Kinnersly-Taylor, J. (2003). <i>Dyeing and Screen-printing on Textiles</i>. London: A & C Black. Miles, L.W.C. (2003). <i>Textile Printing</i>. Bradford: The Society of Dyers and Colourists. Schindler, W.D. (2004). <i>Chemical Finishing of Textiles</i>. Cambridge: Woodhead Publishing in association with the Textile Institut; Boca Raton, Fla.: CRC Press. Heywood, D. (2003). <i>Textile Finishing</i>. Bradford: The Society of Dyers and Colourists.
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